Q-CLUSTER MINI

QUACADEMY
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# Q-Cluster Mini: Your Personal Quantum Computer

*Bringing the power of quantum computing into your hands.*

Q-Cluster Mini is a compact, single-user quantum simulation hardware system that delivers a personal, portable quantum computing experience. Designed as a self-contained device, it enables users to learn, build, and test quantum algorithms anytime with just a local connection to laptop or personal computer. Supporting up to 20 qubits (statevector) and 8 qubits (noise-enabled), it combines foundational learning with realistic experimentation.

**Q-Cluster Mini brings the power of quantum computing into your hands**
**No expensive infrastructure or cloud dependency required.**

Explore a curated set of 10 essential quantum algorithms, modify them, and run simulations seamlessly in an intuitive environment designed for both learning and research. With built-in per-gate noise modelling, Q-Cluster Mini lets you study real-world quantum behaviour by simulating how errors affect circuits and performance—bridging the gap between theory and practical quantum computing.

## Key Features:

- **Personal Quantum Device:** A single-user, portable hardware system; accessible anytime, anywhere.
- **Plug, Connect, Compute:** Connectivity to laptop or personal computer, eliminating dependency on complex infrastructure.
- **High-Fidelity Simulation:** Supports up to 20 qubits (statevector) and 8 qubits (noise-enabled) for realistic experimentation.
- **Per-Gate Noise Modelling:** Analyze how errors impact individual quantum gates and overall algorithm performance.
- **Real Hardware Approximation:** Simulate and study behaviour inspired by real QPUs, including systems from IBM.
- **Built-in Algorithm Library:** Explore 10 core quantum algorithms, from fundamentals to advanced concepts.
- Fully Portable & Self-Contained: Compact design enabling true mobility and personal usage

## Technical Specifications

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| **Device Type** | Personal device (single-user), portable and location-independent quantum simulation hardware box |
| **Connectivity** | Direct local connectivity to system (laptop/personal computer) |
| **Simulation Capacity & Modes** | Up to 20 qubits (statevector simulation)<br>Up to 8 qubits (noise-enabled simulation) |
| **Noise Models** | Gate errors, decoherence, readout noise |
| **Educational Mode** | Circuit design tasks, Quantum algorithms tutorials, Guided modules for approximation of QPU behaviour |
| **Supported Algorithms** | Complete spectrum: Grover's, QFT, Shor's, Simon's, Deutsch-Josza, and many more |
| **User Interface** | Built-in graphical interface for circuit design, execution, and visualization |
| **Output** | Probability graph distribution, Basis state probability distribution, qubit representation on Bloch Sphere, analysis of Bloch sphere, and intuitive interpretation of quantum algorithms |
| **Deployment** | Plug-and-play |

## Learning Outcomes

- Circuit Design & Analysis
- Quantum Algorithm Mastery
- Realistic Quantum Understanding
- Noise Impact Awareness
- Application Prototyping
- Independent Quantum Exploration




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